

GAP ANALYSIS · PROBLEM STATEMENT VS WHAT WE BUILT

Are we building **exactly** what Maharashtra asked for?

Every line of PS 26128 is checked against the actual code in our repository, not our slides or our memory. We also compare it with the systems Maharashtra's animal-husbandry department already runs. The goal is to show the team where we're behind and what to build next.

6**Fully met**

Built, working, demoable

13**Partly met**

Present, but shallow or incomplete

3**Not met**All three are PS *outcomes* we can't measure**57%****Weighted coverage**

Met = 1, Partly = 1/2, over 22 PS items

Verdict: we built the right *kind* of system, and several parts go further than the PS asks for. We're behind on three things: **(1)** we don't measure the outcomes the PS lists, **(2)** we look like a rival to 1962 and Bharat Pashudhan when we should look like the layer that joins them up, and **(3)** alerts reach farmers only inside the app. All three can be fixed before the finale.

Right direction. Three strategic gaps. Five fixes that matter most.

Every capability the PS lists is at least partly there. The judges from the department will probe three places. If we fix those, most of the "partly met" items become "met" as a side effect.

✓ What we got right

- **Real outbreak detection:** a space-time scan statistic that gives a p-value, not a colour-coded guess.
- **Closed loop:** report → triage → sample → lab → cluster → task → compensation.
- **Officials' side** is strong: radar, what-if forecast, SITREP, history, live sync, CSV export.
- **Low-friction reporting:** voice in Hindi/Marathi, icons, offline queue.
- **Food safety:** milk/meat withdrawal alerts after treatment. The PS doesn't ask for this; judges will like it.

⚠ The three strategic gaps

1. **Outcomes are unmeasured.** The PS names 6 expected outcomes. We measure 1. The report form never asks when the animal fell ill (the API hard-codes `onset_date = today`), so we can't show reduced reporting time.
2. **We duplicate Maharashtra's systems.** Its 1962 service already matches symptoms against **108 diseases** (we have 17), and the 1962 Farmer App already holds every animal's history. A judge from the department will spot the overlap straight away.
3. **Advisories stay inside the app.** There's no SMS, push, or voice broadcast. Outbreak advisories also reach only the epicentre village, not the whole zone.

🎯 Five fixes, in order

1. **Onset date + Outcome Scoreboard:** live KPIs for all 6 PS outcomes. *Small.*
2. **Reposition + 1962 Signal Bridge:** take 1962 call tickets and Bharat Pashudhan records as inputs. *Medium.*
3. **Zone broadcast with acknowledgement:** SMS/voice to every village in the risk zone. *Medium.*
4. **LSD-2022 replay:** show on the real Maharashtra outbreak timeline that we detect earlier. *Medium.*
5. **Herd/flock mode + death report:** a one-tap mortality report and flock-level poultry records. *Medium.*

The one sentence to change in our pitch. Stop saying "a new app for farmers and vets." Say: "PashuAarogya is the surveillance brain that sits on top of 1962 and Bharat Pashudhan. It turns their one-animal-at-a-time treatment records into village-level early warning." That answers "why not use 1962?" before anyone asks it.

The eight capabilities the PS lists: what we built vs what they asked for

PS ASKS FOR	WHAT WE ACTUALLY BUILT	STATUS	WHAT'S MISSING	FIX
1 · <u>Capture symptom and mortality reports from farmers and field workers</u>	3-step report: species → symptoms (icons or voice) → count, deaths, photo. Offline queue with de-duplication. Field-worker login tags the channel as field. <code>farmer.js</code> · <code>main.py:283 /api/reports</code>	PARTLY	No onset date, so the API hard-codes today (<code>main.py:308</code>). Mortality is only a number inside the symptom form, with no fast "animal died" path. Field workers see the farmer screen: they can't report on another farmer's behalf or log a door-to-door visit with a "nil report".	Add a "कब से बीमार?" onset picker. Add a one-tap 🏠 death report. Add a field-worker mode: pick farmer/village, nil report.
2 · <u>Rule-based or AI-assisted triage to flag suspected outbreaks</u>	Hot-reloadable YAML rules (17 diseases, 23 signs) → severity band, suspected diseases and a route. Kulldorff-style space-time scan with Poisson p-values finds clusters and auto-creates tasks. <code>engine.py triage()</code> · <code>engine.py scan</code> · <code>rules/diseases.yaml</code>	MET	17 diseases against the 108 that Maharashtra's 1962 system already classifies. Triage accuracy has never been measured, and the scan has only ever run on synthetic data.	Expand the KB to the notifiable-disease list. Back-test on LSD 2022 (innovation C).
3 · <u>Geospatial risk mapping, weather and historical disease trends</u>	Village risk map with an explainable 0–100 score. Live Open-Meteo weather feeds the score. 24-month history by area, disease, species and breed. <code>engine.py:255 weights</code> · <code>weather.py</code> · <code>/api/history</code>	MET	Weather uses <i>current</i> conditions only and one generic rule (humidity ≥ 70%, 22–35 °C) for every disease. History is seeded, not real. No livestock-density, market or vector layers.	Use the 7-day forecast and disease-specific weather rules. Import real historical data where available.
4 · <u>Animal-level or herd-level health, vaccination and treatment records</u>	Per-animal tag, health timeline (vaccinations, cases, treatments, withdrawal), QR Health Passport. <code>/api/animals/{id}/health</code> · <code>passport.html</code>	PARTLY	No herd or flock entity, although poultry and goats aren't individually tagged in practice. No deworming or breeding records. We re-register animals that already have a Pashu Aadhaar in Bharat Pashudhan.	Add a Herd model (counts by species/age). Key everything on the 12-digit Pashu Aadhaar. Read records from NDLM instead of re-typing them.

Our dashboards are strong. Reaching farmers outside the app is our weakest area.

PS ASKS FOR	WHAT WE ACTUALLY BUILT	STATUS	WHAT'S MISSING	FIX
5 • Issue multilingual advisories and alerts	Advisories auto-generated in मराठी / हिन्दी / English when a cluster is detected. Treatment withdrawal alerts and camp notices. The assistant reads answers aloud. engine.py:234 • main.py:427	PARTLY	In-app only: a farmer who doesn't open the app never sees them. No push, SMS, WhatsApp or voice broadcast. Bug: the outbreak advisory goes to the epicentre village only (<code>village_id=center.id</code> , <code>engine.py:240</code>), although the comment says "zone villages". No read or acknowledgement tracking.	Send to every village in <code>zone_village_ids</code> . Add web push and an SMS gateway. Add "press 1 once done" acknowledgement (innovation E).
6 • Sample collection, laboratory referral and case escalation	Sample with a chain-of-custody code and 5 stages (COLLECTED → RESULT). A positive result auto-confirms the case and alerts officials. Escalation goes block → district → state. main.py:449 • :485 • :398 <code>escalate</code>	PARTLY	There's one generic "lab": no lab directory and no routing by disease (avian influenza must go to a BSL-3 national lab). No guidance on sample type or cold chain. Turnaround time isn't tracked. There's no way to refer the <i>animal</i> to a polyclinic.	Add a lab directory with route-by-suspected-disease, a turnaround-time clock, and a "refer to polyclinic" action (innovation G).
7 • Dashboards for veterinary officials	Vet triage queue and lab board. Block/district/state command centre with radar, what-if forecast, action queue, claims, camps, history, SITREP, CSV export and live cross-dashboard sync. gov.html • vet.html • sitrep.html	MET	The dashboards show <i>activity</i> , not <i>performance</i> : no per-block response times, no pending work per officer, no targets.	Add an Outcome Scoreboard tab with per-block league tables (innovation B).
8 • Mobile, web, IVR or offline-enabled channels	Responsive web and an installable PWA. Offline report queue that syncs on reconnect. Voice assistant. IVR simulator page. sw.js • api.js <code>queue</code> • ivr.html	PARTLY	The IVR is a browser simulator , not a phone number anyone can call. The <code>sms</code> channel exists as a label only. The OTP is fixed at <code>123456</code> and no SMS is ever sent (<code>main.py:134</code>).	Use a trial Exotel/Twilio number: missed call → Marathi IVR callback. Parse structured SMS such as <code>LSD 3 1</code> (innovation D).

The PS promises six outcomes. We can currently show one.

This is our biggest hole, and it's cheap to fix. Judges score *impact*, and impact means numbers that move. Most of the timestamps we need are already in the database. We just never compute the durations.

EXPECTED OUTCOME	STATUS	WHERE WE STAND	THE KPI TO SHOW (AND WHERE THE DATA ALREADY LIVES)
Reduced reporting time	NOT MET	Can't be measured: <code>onset_date</code> is hard-coded to today (main.py:308) and the seeded data sets <code>onset</code> = report date. The delay is always zero .	Median onset → report, in hours , trended weekly and split by channel (app / field / IVR). Needs the onset picker first.
Earlier outbreak identification	NOT MET	We detect clusters, but we claim "earlier" without evidence. There's no baseline to compare against.	Detection lead-time: <code>Outbreak.detected_at</code> vs the first lab-confirmed case in the zone. Plus the LSD-2022 replay (innovation C).
Improved vaccination coverage	PARTLY	An overall 12-month coverage % is shown (main.py:538), and camps can be scheduled.	Coverage vs target per village, due and overdue (defaulter) lists, and coverage gained per camp.
Faster treatment and containment	NOT MET	The timestamps exist (<code>reported_at</code> , <code>collected_at</code> , <code>result_at</code> , <code>Task.done_at</code>), but no durations are computed.	Report → first vet action , measured against Maharashtra's own <i>1-hour callback norm</i> . Sample → result turnaround. Cluster → ring vaccination.
Lower mortality and productivity loss	PARTLY	Deaths in the last 7 days are counted (main.py:527), with no rate and no trend.	Case-fatality rate by disease, deaths per 1,000 animals, and an estimated ₹ milk loss per outbreak.
Stronger evidence-based planning	MET	SITREP, the what-if ring-vaccination forecast, historical trends and CSV exports.	Already demoable. Add "decisions taken from this SITREP" to close the loop.

FOUND IN THE CODE: A KPI WE PROMISED OURSELVES AND NEVER BUILT

`main.py:539` has the comment `# median report->awareness lag in hours (reported_at vs onset)` directly above the return statement. The metric was never implemented. It's the first outcome in the PS, and we'd already planned to build it.

Every pain point is addressed. None is fully closed except two.

PAIN POINT NAMED IN THE PS	STATUS	HOW WE ADDRESS IT TODAY	WHAT WOULD CLOSE IT
No real-time risk view at <u>village, block and district</u> level	MET	LGD-coded village → block → district hierarchy. Drill-down, risk bands and live sync across dashboards.	Already strong. Keep it as a headline in the demo.
Symptoms reported <u>late</u>	PARTLY	Voice, icons and offline queueing all lower the friction of reporting.	Prove it: measure the onset → report delay (outcome 1).
Diagnostic facilities are <u>distant</u>	PARTLY	MVU dispatch tasks are auto-created for each cluster.	Nearest-lab and polyclinic routing, sample transport, a turnaround-time clock.
Vaccination and treatment histories <u>incomplete</u>	PARTLY	Full history for animals registered in <i>our</i> app.	Pull histories from Bharat Pashudhan by Pashu Aadhaar, so we complete the record instead of starting a new one.
Information from <u>farms, dispensaries, labs, vaccination drives and surveillance programmes</u> is fragmented	PARTLY	Unified, but only for data typed into our app. There's no dispensary entity and no connector to 1962, NDLM or NADRES.	This is the core complaint of the PS. Build the 1962 Signal Bridge (innovation A) plus a dispensary OPD import.
Risk of <u>zoonotic</u> transmission	PARTLY	A One Health flag on anthrax, rabies, brucellosis and avian-flu clusters, plus an alert to <code>target_role="health"</code> .	No user can hold a <code>health</code> role, so nobody in public health ever reads the alert. Add a One Health desk (innovation H).
Mortality and productivity loss hit <u>farmers' incomes</u>	MET	Compensation-claim loop (report → claim → officer approval → farmer sees it). Withdrawal alerts protect milk sales.	Add DBT payout status and ₹ loss estimates.
Must work in <u>low-connectivity areas</u>	PARTLY	Offline-first PWA with a sync queue.	A real phone channel (missed call, IVR, SMS) for farmers with feature phones.

Don't compete with 1962 and Bharat Pashudhan. Sit on top of them.

EXISTING SYSTEM	WHAT IT ALREADY DOES	OVERLAPS WITH US	WHERE WE MUST DIFFER
1962 helpline + MVUs <i>Maharashtra AHD + BFIL</i> <i>· CM Animal Health Scheme</i>	Call centre at the AH Commissionerate, Pune. Symptoms are matched against 108 diseases by severity and maximum response time. A vet must call the owner within 1 hour . 800,000+ animals treated in 5 years.	Our symptom triage, MVU dispatch	It handles one animal at a time . Nothing we found aggregates those calls into clusters. We ingest them.
Bharat Pashudhan / NDLM <i>DAHD + NDDB · 1962 Farmer App</i>	12-digit Pashu Aadhaar ear tag. Lifetime records of vaccination, treatment, breeding and tests. E-prescriptions.	Our animal registry, passport, vaccination records	Read from it by Pashu Aadhaar. Never ask a farmer to register an animal twice.
NADRES v2 <i>ICAR-NIVEDI, Bengaluru</i>	AI forecasts for 13 economically important diseases, at district level, as monthly bulletins using GIS, weather and remote-sensing layers.	Our forecast and risk map	Ours is village-level and case-driven , and updates with every report. Use NADRES as a prior in our risk score.
WhatsApp chatbot <i>AHD + BFIL</i>	Disease information, vaccination schedules, video consultations. In Raigad: 615 farmers used it and 201 video consultations were held (Oct 2025 – May 2026).	पशु मित्र assistant	Voice-first in Hindi and Marathi, grounded in live local outbreak data, and it acts : it files the report itself.

ACTION

Advisories to the zone · ring vaccination · movement permits · compensation · SITREP

▲

PASHU AAROGYA

The surveillance brain. Cluster detection, risk, forecast, tasks and outcome KPIs, across *all* sources

▲ ▲ ▲

SIGNALS IN

1962 call tickets · Bharat Pashudhan treatments and vaccinations · lab results · dispensary OPD · farmer and field-worker reports · NADRES forecasts

Why this wins. The PS complains that information "may remain fragmented". A new standalone app adds one more fragment. A layer that *joins* the existing systems is the only answer that doesn't make the problem worse. It's also the only story in which the department doesn't have to retrain 1962 staff.

HONEST RISK

The adapters can be demoed with CSV and mock APIs, but live access to 1962 or NDLM data needs a government MoU. Say that openly, as a Phase 1 dependency.

Ten things we should lead the demo with, because nobody else will have them

STRENGTH	WHY IT LANDS WITH JUDGES
Space-time scan statistic	A peer-reviewed epidemiology method with a p-value. It turns "we think" into "p = 0.003".
What-if ring vaccination	SEIR model on the village graph. Shows "cases prevented" and doses needed, which is exactly the planning the PS asks for.
Reporting pays	Report → claim → approval. It fixes the <i>incentive</i> behind under-reporting, not just the interface.
Explainable risk score	Tap a village to see why it's high: burden, growth, history, vaccination gap, cluster, weather.
Food-safety withdrawal	Milk/meat withdrawal countdown after treatment, in three languages. Links animal health to human health.

STRENGTH	WHY IT LANDS WITH JUDGES
Movement permit + passport	Scan a QR code and see "BLOCKED: inside LSD zone". Containment enforced at the checkpoint.
Offline queue with dedup	Switch to airplane mode, file a report, and it syncs once. Each report has a UUID, so nothing is counted twice.
Grounded voice assistant	The rules layer answers instantly offline. Gemini handles free-form Hindi and Marathi. It never gives a diagnosis or a drug dose.
Auto SITREP	The morning one-pager a District Animal Husbandry Officer would otherwise compile by hand.
Hot-reload knowledge base	"Add a disease" is 20 lines of YAML and one click. The department owns the rules, not us.

About Pashu Lens (breed ID). It's impressive, but breed identification isn't in the PS. If we lead with it, a department judge may ask "how does breed ID detect disease?" Present it second: it speeds up registration and enables breed-wise risk analysis. Point out that the same vision pipeline could be retrained for lesion screening (innovation J), which is in scope.

Six additions that turn "partly met" into a wow moment

A 1962 Signal Bridge

Import 1962 call-centre tickets and Bharat Pashudhan treatment records as a surveillance feed. Every call becomes a data point on the radar. Match animals by Pashu Aadhaar.

DEMO ▶ Upload a CSV of 30 "routine" 1962 calls. The radar lights up a cluster that nobody had noticed.

PS: fragmented information Effort M ★★★

B Outcome Scoreboard

Live KPIs for all six PS outcomes: onset → report, report → vet contact (against the state's 1-hour norm), lab turnaround, detection lead-time, coverage vs target, case-fatality rate. Includes a per-block league table.

DEMO ▶ Each PS outcome gets a live number and a trend, e.g. "median reporting delay: X days, down from Y". Illustrative wording only, never quote invented figures.

PS: expected outcomes Effort S-M ★★★

C LSD-2022 Replay

Replay Maharashtra's real 2022 lumpy skin disease timeline through our radar. Officially reported deaths climbed from 126 (18 Sep 2022) to about 34,700. Show when our scan would have raised the alarm.

DEMO ▶ A time-slider over Jalgaon, Ahmednagar, Akola and Pune: "flagged N days before the official count."

PS: earlier identification Effort M ★★★

D Missed-call reporting

A farmer gives a missed call to one number and gets a free callback in a Marathi IVR, and a case is filed. Also accept structured SMS such as LSD 3 1 (disease, sick, dead). Works on any feature phone.

DEMO ▶ A judge gives a missed call from their own phone. The case appears on the dashboard.

PS: IVR, low connectivity Effort M ★★★

E Zone broadcast + acknowledgement

When a cluster is detected, every household in the risk zone gets an SMS or voice call in its own language: "Press 1 once you've isolated or vaccinated." The map then shows compliance, not just risk.

DEMO ▶ Villages turn from red to green as farmers acknowledge.

PS: advisories, containment Effort M ★★

F Herd mode + mass-mortality trigger

Flock and herd records for poultry and goats. If at least X% of a flock dies within 48 hours, the case auto-escalates as suspected avian flu: biosecurity protocol, BSL-3 lab routing and a One Health notice.

DEMO ▶ Report "40 of 300 birds dead". The whole zoonotic chain fires in one step.

PS: herd records, mortality Effort M ★★

Four more that deepen our coverage of the PS

G Smart lab referral

A lab directory (district lab → regional lab → national BSL-3). Route each sample by suspected disease and distance, with a checklist for sample type and cold chain, and a turnaround-time clock that escalates if the result is late.

DEMO ▶ A suspected avian-flu sample is auto-routed to the national BSL-3 lab, with a packing checklist and an ETA.

PS: laboratory referral Effort S-M ★★

H One Health desk

A real login for the district health officer (IDSP), who receives zoonotic notices with an acknowledgement deadline and can report human cases back. Today our health alerts go to a role that no user can have.

DEMO ▶ An anthrax cluster in cattle leads the health officer to acknowledge and schedule a human-surveillance visit.

PS: zoonotic risk Effort S ★★

I Vaccination gap map + defaulter tracing

Show coverage against a herd-immunity target for each village. Give field workers due and overdue lists, and send reminders automatically. Ring-vaccination plan → camp → coverage gained, measured end to end.

DEMO ▶ e.g. "This village is at N% against a 70% target: M animals due. Camp booked, reminders sent."

PS: vaccination coverage Effort M ★★

J Lesion-photo screening (stretch)

Retrain the existing Pashu Lens pipeline to spot skin and mouth lesions: LSD nodules, FMD vesicles. The output raises suspicion only and a vet confirms. The risk is getting enough labelled images; frame it as a research track.

DEMO ▶ A photo of a nodular cow → "LSD suspicion ↑, confirmation needed", with the triage band raised.

PS: AI-assisted triage Effort L ★★★ wow / ⚠ risk

How to pick. A, B and C are the three that change the judges' verdict: they answer "is it new?", "does it work?" and "can you prove it?". D is the most memorable live moment. The rest deepen coverage. If time is short, build A-D and describe E-J on a slide.

Work plan: fix the cheap gaps first, then build the differentiators

PRIO	TASK	WHERE IN THE CODE	EFFORT	DONE WHEN...	OWNER
P0	Onset-date picker + reporting-delay KPI	farmer.js report form · ReportIn + main.py:308	S	A new report stores the real onset date and the delay shows on the dashboard	_____
P0	Send advisories to the whole zone (bug)	engine.py:235-241 → loop over zone_village_ids	S	Every village in the cluster radius gets the advisory	_____
P0	Outcome Scoreboard (innovation B)	main.py:518 summary · new tab in gov.js	M	All 6 PS outcomes show a number and a trend	_____
P0	Rewrite the pitch: "layer on top of 1962 / Bharat Pashudhan"	Slides, README.md, login page tagline	S	Every teammate can say the sentence on page 2 from memory	_____
P0	One-tap death report	farmer.js · triage already handles dead_count	S	A death is reported in 2 taps and triaged as high	_____
P1	1962 Signal Bridge (A)	new backend/integrations.py · CSV + mock API	M	Imported tickets appear as cases and can form a cluster	_____
P1	LSD-2022 Replay (C)	new replay script + time slider on the radar	M	Slide: "flagged N days before the official count"	_____
P1	Missed call / IVR / SMS (D)	gateway webhook in main.py · reuse /api/reports	M	A real phone call files a real case	_____
P1	Herd mode + mortality trigger (F)	models.py Herd · engine.py rule	M	A mass poultry death fires the One Health chain	_____
P1	Lab directory + turnaround time (G), One Health desk (H)	models.py Sample.lab_id · new health role	M	Samples route by disease, and the health officer can log in	_____
P2	Grow the KB toward the full notifiable list · automated tests for triage and scan · real OTP and a DPDP-Act privacy note	rules/diseases.yaml · new tests/	L	≥ 40 diseases · a CI badge · consent screen	_____

The questions a department judge will ask, and how to answer them

LIKELY QUESTION	RISKY ANSWER TODAY	STRONG ANSWER (AFTER THE FIXES)
"We already have 1962 and Bharat Pashudhan. Why yours?"	"Ours has more features."	"They treat one animal at a time. We're the layer that turns their records into village-level early warning, and we read their data by Pashu Aadhaar."
"How do you know reporting time goes down?"	We can't: onset isn't captured.	Show the median onset → report KPI on the Outcome Scoreboard.
"Is this data real?"	"Yes." (It's synthetic. Never say this.)	"It's synthetic, shaped like a real LSD wave. Here's our replay on the public Maharashtra 2022 timeline."
"How accurate is the triage?"	A percentage we never measured.	"Triage only routes, it never diagnoses. A vet confirms every case, and the confirmation rate is tracked. The department edits the rules."
"What about a farmer with no smartphone?"	"The IVR page" (a simulator).	A live missed call from the judge's phone, plus SMS codes and field-worker proxy reporting.
"Why breed identification?"	Leading with it as the main AI.	"It speeds up registration. The same pipeline is being retrained for lesion screening."
"Privacy? Who owns the data?"	No answer prepared.	"Consent at sign-up under the DPDP Act, role-scoped access, minimal personal data. The department owns the data."

Sources

- Maharashtra DAHD: Chief Minister Animal Health Scheme (108 diseases, 1-hour callback, Pune call centre): dahd.maharashtra.gov.in
- Free Press Journal, 2 Jun 2026: 1962 helpline and WhatsApp chatbot in Raigad: freepressjournal.in
- Bharat Pashudhan / NDLM overview: olrds.odisha.gov.in/en/ndlm · 1962 Farmer App: [Google Play](#)
- ICAR-NIVEDI NADRES v2: nivedi.res.in/Nadres_v2
- LSD in Maharashtra 2022 (126 deaths, 18 Sep): indiatvnews.com · cumulative toll: [Wikipedia](#)
- SIH judging criteria (novelty, feasibility, impact, UX, scalability): whereuelevate.com

Method. A code audit of the repository at `a63324a`. Each status is traced to a file and line. The 22 PS items are: 8 capabilities, 6 outcomes and 8 pain points. Weighted coverage counts Met as 1 and Partly as ½.